

UTSJOKI KEVO, Finland

WMO: 028050

Lat: 69.756N

Lon: 27.007E

Elev: 349

StdP: 14.51

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.3	(c) -24.8	(d) -36.0	(e) 0.7	(f) -29.9	(g) -30.2	(h) 1.0	(i) -24.6	(j) 30.4	(k) 25.1	(l) 24.1	(m) 24.6	(n) 3.1	(o) 160	(p) 0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	75.8	61.2	71.3	59.4	67.0	57.3	63.1	72.4	60.6	69.1	58.4	65.7	7.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.3	76.4	66.7	56.9	69.9	63.5	54.7	64.5	61.3	28.8	72.3	26.9	69.3	25.5	66.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.8	(b) 18.7	(c) 15.7	(e) -35.0	(f) 82.4	(g) 6.0	(h) 4.2	(i) -39.3	(j) 85.4	(k) -42.8	(l) 87.9	(m) -46.1	(n) 90.2	(o) -50.5	(p) 93.3		
(a) 22.8	(b) 18.7	(c) 15.7	(e) -35.0	(f) 66.2	(g) 5.9	(h) 2.8	(i) -39.3	(j) 68.2	(k) -42.7	(l) 69.8	(m) -46.0	(n) 71.4	(o) -50.3	(p) 73.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.3	7.1	8.0	17.1	28.0	39.1	49.2	56.2	52.1	43.8	31.6	18.0	12.3		
	DBStd	20.07	16.14	15.54	11.99	9.10	6.86	6.75	6.54	6.11	6.16	8.84	12.63	14.71		
	HDD50	7588	1329	1175	1020	661	347	94	15	46	202	572	959	1168		
	HDD65	12661	1794	1595	1485	1111	803	474	282	402	636	1037	1409	1633		
	CDD50	415	0	0	0	0	9	71	208	111	16	0	0	0		
	CDD65	14	0	0	0	0	0	1	10	3	0	0	0	0		
	CDH74	215	0	0	0	0	6	19	150	40	0	0	0	0		
	CDH80	41	0	0	0	0	2	2	32	5	0	0	0	0		
	WSAvg	6.5	6.1	5.8	6.8	6.6	7.3	8.3	7.3	6.1	6.2	6.0	5.6	6.1		
	PrecAvg	16.4	1.0	0.8	0.8	0.8	0.9	1.8	2.7	2.3	1.6	1.5	1.1	1.0		
(15) Precipitation	PrecMax	23.5	2.6	2.6	1.6	2.2	3.2	4.5	6.2	6.3	3.2	3.8	2.4	2.1		
	PrecMin	10.4	0.2	0.1	0.2	0.2	0.0	0.1	0.5	0.4	0.2	0.3	0.3	0.3		
	PrecStd	3.3	0.5	0.5	0.4	0.4	0.6	1.0	1.4	1.1	0.7	0.8	0.5	0.4		
	PrecStd	3.3	0.5	0.5	0.4	0.4	0.6	1.0	1.4	1.1	0.7	0.8	0.5	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	38.5	41.0	51.8	70.9	76.7	84.1	79.2	64.6	51.7	40.1	40.0		
		MCWB	35.0	35.6	36.6	43.1	53.8	59.7	64.1	63.6	55.6	48.1	38.5	36.6		
	2%	DB	34.0	35.3	37.6	46.5	61.9	71.2	78.0	72.5	60.2	48.8	36.8	35.7		
		MCWB	32.1	33.0	34.3	40.0	49.5	56.7	62.7	60.9	54.0	45.9	35.1	33.6		
	5%	DB	30.6	32.1	34.7	42.6	55.7	66.0	73.3	67.4	56.7	45.4	35.0	33.2		
		MCWB	29.1	30.3	31.8	37.4	46.6	54.4	60.4	58.5	52.1	43.2	33.8	31.7		
	10%	DB	27.6	29.0	32.3	39.7	50.6	61.6	69.0	63.1	53.7	42.4	33.2	31.0		
		MCWB	26.4	27.4	29.8	35.5	43.6	52.3	59.2	56.4	50.1	40.5	32.2	29.7		
	0.4%	WB	35.3	36.2	37.0	43.7	55.9	61.8	67.0	65.4	57.7	48.7	38.7	36.9		
		MCDB	37.3	38.3	40.5	50.8	69.3	71.3	77.7	74.8	62.9	51.0	39.9	39.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.1	33.1	34.4	40.3	50.5	58.6	64.4	62.0	54.9	46.0	35.6	34.0		
		MCDB	33.8	35.2	37.1	46.1	60.8	67.5	74.2	70.7	58.7	48.3	36.4	35.5		
	5%	WB	29.3	30.4	32.1	37.7	47.1	55.8	62.0	59.2	52.7	43.3	34.0	31.9		
		MCDB	30.6	32.2	34.4	41.9	54.7	64.1	71.2	66.1	56.2	45.3	34.9	33.2		
	10%	WB	26.4	27.4	29.9	35.9	44.2	53.1	59.6	57.1	50.5	40.8	32.3	29.6		
		MCDB	27.6	28.9	32.3	39.2	50.1	60.8	67.9	62.5	53.5	42.4	33.2	31.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.9	16.1	18.3	17.1	14.4	14.9	16.2	15.7	13.2	9.4	12.6	15.1		
		MCDBR	13.6	12.2	12.9	17.0	25.3	23.7	25.9	24.6	18.3	10.6	9.3	11.8		
		MCWBR	12.8	11.4	10.9	11.6	14.6	11.7	12.3	13.2	12.2	8.5	8.7	11.2		
		MCDBR	13.5	11.8	12.9	15.8	23.2	20.9	22.9	21.3	16.4	10.1	9.0	11.7		
	5% WB	MCWBR	12.9	11.0	11.1	11.0	14.0	11.1	11.9	12.1	11.4	8.4	8.5	11.2		
		MCWBR	12.9	11.0	11.1	11.0	14.0	11.1	11.9	12.1	11.4	8.4	8.5	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.202	0.247	0.295	0.324	0.346	0.344	0.356	0.338	0.306	0.260	N/A	N/A		
	taud		1.903	2.027	2.104	2.156	2.314	2.452	2.444	2.475	2.525	2.206	N/A	N/A		
	Ebn at Noon		4	148	220	254	264	270	261	253	231	146	0	0		
	Edh at Noon		1	13	25	33	32	30	29	25	18	11	0	0		
(44) All-Sky Solar Radiation	RadAvg		3	111	542	1134	1485	1502	1387	957	499	151	0	0		
	RadStd		0	10	35	68	154	135	152	91	44	17	0	0		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page